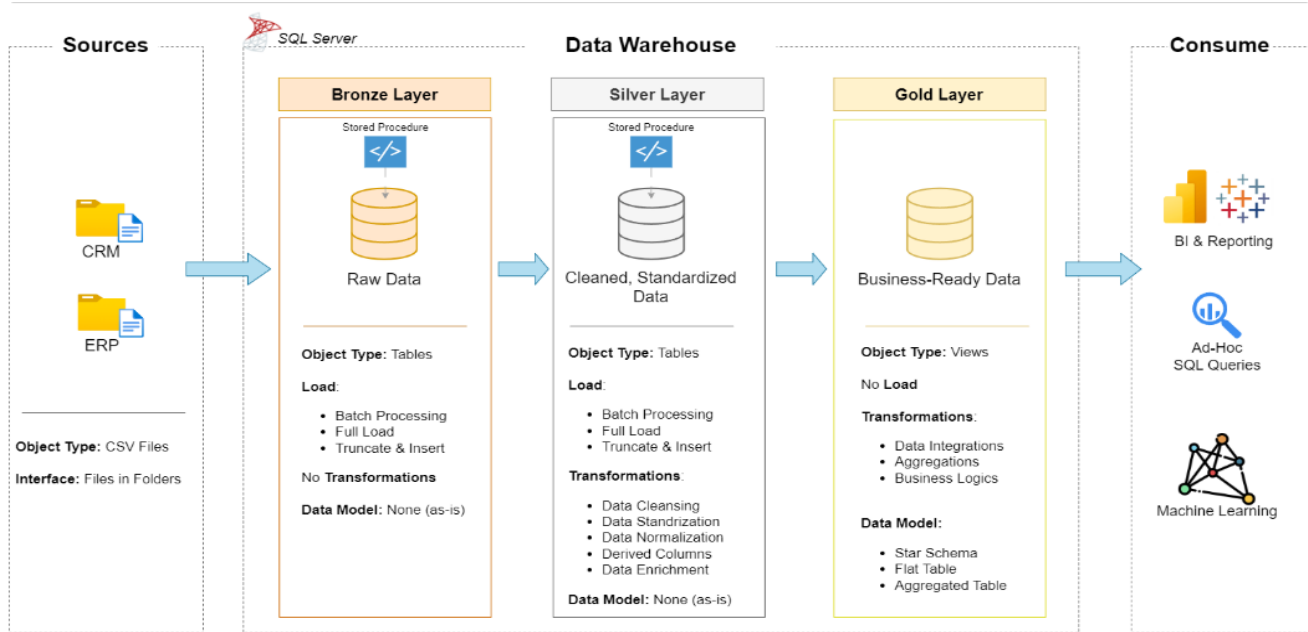


■ Advanced Data Warehouse Architecture (Bronze–Silver–Gold)



■ Bronze Layer - Detailed Flow

Step	Description
Ingestion	Data loaded from CRM, ERP, CSV
Storage	Stored as raw tables
Processing	Batch / Full Load
Transformation	None (as-is)

```
SELECT * FROM sales_raw;
```

■ Silver Layer - Detailed Flow

Step	Description
Cleaning	Remove nulls, duplicates
Standardization	Format & naming consistency
Normalization	Structured schema
Enrichment	Derived columns, joins

```
SELECT
customer_id,
TRIM(customer_name),
COALESCE(sales,0)
FROM sales_clean;
```

■ Gold Layer - Detailed Flow

Step	Description
Integration	Combine multiple datasets
Aggregation	SUM, COUNT, KPIs
Modeling	Star schema / Flat tables
Output	Views for BI tools

```
SELECT product_id,
SUM(sales) total_sales
FROM sales_gold
GROUP BY product_id;
```

■ Consumption Layer Flow

Tool	Usage
Power BI/Tableau	Dashboards
SQL	Ad-hoc queries
ML Models	Predictions

■ Advanced Benefits

- Scalability for large datasets
- Clear separation of concerns
- Faster query performance
- Supports ML pipelines and forecasting systems